Benchmarking deep learning against matched filtering for single-event graviton-dispersion tests on LIGO data

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Abstract

General relativity predicts that the graviton—the carrier of the gravitational force—is exactly massless; a broad class of theories beyond general relativity instead gives it a small rest mass m_g. A nonzero m_g would make gravitational waves *dispersive*, imprinting a calculable, frequency-dependent phase on a binary-black-hole inspiral [1, 2], so that measuring this phase tests whether the graviton has mass. To carry out this test we employ a deep-learning approach and benchmark it against the optimal matched filter on a *single* gravitational-wave event. This is a demanding regime: the strongest current bound, $\lambda_g = h/(m_g c) > 1.4 \times 10^{16}\,\mathrm{km}$, is reached only by stacking $\mathcal{O}(20)$ events [3], because any single event is buried in non-Gaussian detector noise, and whether a learned model can outperform matched filtering on one event has not previously been tested on observational strain. We inject general-relativistic and massive-graviton inspirals—source parameters drawn from the GWTC-4.0 catalogue, dispersion phase to leading order in $1/\lambda_g^2$—into LIGO H1 strain near GW150914 at controlled signal-to-noise ratios $\rho_{\mathrm{opt}} \in \{20, 50\}$, scan the graviton Compton wavelength λ_g, and on identical data compare two single-event discriminators: the *oracle* matched filter, handed the true source parameters, and a compact (1.7×10^5-parameter) convolutional–transformer classifier, given only whitened strain. On observed strain both reach only an area under the receiver–operator curve of AUC = 0.45–0.49, statistically indistinguishable from each other and from chance at every grid cell, while two controls confirm the implementation is correct—the published O3 bound is reproduced, and on noise-free signals the matched filter separates the two classes perfectly (AUC = 1). The single-event ceiling is therefore set by transient "glitch" noise, not by the choice of discriminator: at present-day signal-to-noise ratios deep learning offers no per-event shortcut around the noise floor that forces event stacking. Our contribution is this validated, openly reproducible benchmark—identical injections, an oracle matched-filter baseline, and a like-for-like learned comparison—which quantifies the single-event ceiling and sets a baseline against which any future claim of a learned single-event advantage must be measured. All code, the curated event list, and the result data are available at `https://github.com/ramc77/gw-graviton-ml-benchmark`.

1 Introduction

Gravitational waves—propagating ripples in spacetime—are produced when two black holes orbit one another, lose orbital energy to that radiation, and spiral inward until they merge. The signal is a brief "chirp" that sweeps upward in frequency over the final fraction of a second and is recorded on Earth by the kilometre-scale laser interferometers of LIGO and its partner observatories Virgo and KAGRA [4, 5], which sense the minute stretching and squeezing of

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space the passing wave produces—the *strain*. Since the first such detection, the binary-black-hole merger GW150914, dozens of black-hole mergers have been catalogued, and because each chirp spans a broad band of frequencies it is a sensitive probe of how gravity itself propagates—in particular, of the dispersion relation obeyed by the gravitational field.

In general relativity the graviton—the spin-2 quantum of the gravitational field—is exactly *massless*, so a gravitational wave of every frequency travels at the speed of light and the chirp reaches the detector undistorted. The graviton mass m_g is therefore a sharp parameter with which to test gravity: any nonzero rest mass modifies the gravitational-wave (GW) dispersion relation, so that different frequencies propagate at slightly different speeds and a calculable, frequency-dependent phase shift accumulates on the observed binary-black-hole inspiral waveform [1, 2]. Limits on m_g have historically been set from solar-system dynamics and a range of astrophysical and laboratory probes [6, 7, 8], but the tightest current values come from gravitational-wave inspirals, in which the dispersion accumulates over cosmological propagation distances [9]. From this growing catalogue of black-hole mergers the LIGO–Virgo–KAGRA collaboration has constrained m_g through hierarchical analysis of all events jointly [10]; the third-observing-run combination yields a 90% credible upper limit $m_g c^2 < 1.27 \times 10^{-23}\,\mathrm{eV}$, or equivalently $\lambda_g = h/(m_g c) > 1.4 \times 10^{16}\,\mathrm{km}$ [3]. This bound is the catalogue-level result; the corresponding constraint from any single event is typically two orders of magnitude weaker, because single-event matched-filter likelihoods are limited by the variance contributed by non-Gaussian noise transients in the LIGO strain [11, 12].

Deep learning has been widely promoted for gravitational-wave analysis, which raises a sharp methodological question: can a learned model extract more information from a *single* event than the optimal matched filter does? In the GW-signal-versus-noise problem, deep classifiers have been shown to recover the sensitivity of matched filtering at substantially lower inference cost, both on synthetic data [13, 14] and on archival LIGO observations [15]; learned models now also perform full Bayesian parameter estimation in near real time [16]. The corresponding question for the more specific task of distinguishing two waveform *families* (here, general relativistic versus massive-graviton templates with otherwise identical source parameters) has, to our knowledge, not been examined on observational strain data, and it is this benchmarking question—rather than a new bound on m_g—that we set out to answer.

We address that question directly. We construct balanced injection sets in which each example is a 4-second segment of H1 strain near the GW150914 epoch, containing a binary black hole inspiral drawn from GWTC-4.0 source parameters and modified by either a Mirshekari–Yunes dispersion phase [2] or by nothing. Each injection is rescaled to a controlled optimal matched-filter signal-to-noise ratio. We then compare two single-event classifiers on the same examples: (i) the oracle matched filter, which has access to the true source parameters (m_1, m_2, D_L) and constructs the GR and massive-graviton templates at those parameters; and (ii) a 1D convolutional plus transformer classifier with no explicit source-parameter input. We find that the two discriminators are statistically indistinguishable and that both perform no better than chance on a single event—a null result we trace to the non-Gaussian detector noise rather than to the choice of method, and which we confirm with two independent controls. Our central contribution is therefore not a new bound on m_g but a validated, openly reproducible benchmark: a controlled injection set, an oracle matched-filter baseline, and a like-for-like learned comparison on identical data, which together quantify the single-event ceiling and provide a baseline against which future learned methods can be measured.

The paper is organised as follows. Section 2 states the waveform model, the injection procedure, the matched-filter baseline, and the classifier architecture. Section 3 establishes that the implementation reproduces the published O3 bound to within a factor consistent with single-event versus catalogue SNR scaling. Section 4 reports the per-event sensitivity comparison. Section 5 interprets the comparison in the context of the LIGO–Virgo–KAGRA glitch literature and lists explicit limitations of the present study; Sec. 6 states the conclusions.

2 Methods

2.1 Waveform model

We use the `IMRPhenomD` frequency-domain inspiral–merger–ringdown model of Khan et al. [17, 18], as implemented in `PyCBC` [19], as the general-relativistic baseline waveform $\tilde{h}_{\rm GR}(f;\boldsymbol{\theta})$. For the massive-graviton waveform $\tilde{h}_{\rm MG}(f)$ we add to the GR phase the Mirshekari–Yunes modification [2],

$$\delta\Psi_{\rm MG}(f) = -\frac{\pi^2\, c\, D_L}{\lambda_g^2\,(1+z)\,\mathcal{M}_c\, f}, \tag{1}$$

where D_L is the luminosity distance, $\mathcal{M}_c = GM_c/c^3$ is the chirp mass in seconds, and we work to leading order in $1/\lambda_g^2$. For the events considered ($z \lesssim 0.2$) the cosmological distance D_0 appearing in the original derivation reduces to $D_L/(1+z)^2$ at the few-percent level, and we take $D_0 = D_L$ in Eq. (1). The massive-graviton waveform is then

$$\tilde{h}_{\rm MG}(f) \;=\; \tilde{h}_{\rm GR}(f)\, \exp[i\,\delta\Psi_{\rm MG}(f)]\,. \tag{2}$$

We emphasise that $\delta\Psi_{\rm MG} \propto D_L$: when we rescale a template's distance to a different SNR, we must regenerate the dispersion phase at the new distance, not at a fixed reference.

2.2 Injection construction

We draw component masses (m_1, m_2) with replacement from the 56 GWTC-4.0 [20] BBH events with both source-frame masses above $3\,M_\odot$ and network matched-filter SNR above 10 (*cf.* Table 1); the catalogue parameters are fetched directly from the `gwosc` JSON endpoint [21, 22]. For each draw and each target optimal SNR $\rho_{\rm opt}$, we compute a reference GR waveform at $D_L = 100\,\mathrm{Mpc}$, evaluate its optimal SNR on the strain PSD, and rescale the luminosity distance so that the injected optimal SNR equals $\rho_{\rm opt}$ exactly.

The noise component is taken from a 1024-second segment of Advanced LIGO [4] H1 strain ending 16 s before the GW150914 [5] epoch, downloaded via `gwpy.timeseries.TimeSeries.fetch_open_data` [22]. The one-sided power spectral density $S_n(f)$ is estimated by a median-Welch periodogram with 4-second Hann windows and 50% overlap, following [23]. Each 4-second noise segment used in an injection is drawn from a uniform random offset in the reservoir.

For the classifier input we whiten each 4-second segment by $1/\sqrt{S_n(f)}$ in frequency domain, then standardise each segment to zero mean and unit variance over the interior 75% (to remove whitening edge artefacts). The matched-filter analysis operates directly on the un-whitened strain through the PSD-weighted inner product.

2.3 Matched-filter baseline

We adopt the optimal-filter conventions of [23, 24]: for two frequency-domain signals $\tilde{a}, \tilde{b}$ and a one-sided PSD S_n,

$$\langle a|b\rangle \;=\; 4\,\mathrm{Re}\int_{f_{\rm low}}^{f_{\rm high}} \frac{\tilde{a}^*(f)\,\tilde{b}(f)}{S_n(f)}\, df, \tag{3}$$

with $f_{\rm low} = 20\,\mathrm{Hz}$ and $f_{\rm high} = 1024\,\mathrm{Hz}$ throughout. The optimal SNR of a template h in data d is $\rho(h) = \langle d|h\rangle/\sqrt{\langle h|h\rangle}$. For each injection we know the true source parameters (m_1, m_2, D_L), and the oracle matched filter constructs both $\tilde{h}_{\rm GR}$ and $\tilde{h}_{\rm MG}(\lambda_g)$ at these parameters. The discriminator is

$$s_{\rm MF} \;\equiv\; \rho_{\rm MG}^2 - \rho_{\rm GR}^2, \tag{4}$$

which in Gaussian noise is twice the log-likelihood ratio between the massive-graviton and general-relativistic hypotheses [23]. This is the strongest possible matched-filter test of the dispersion hypothesis on a single event: any non-oracle template bank produces a discriminator no more powerful than $s_{\rm MF}$.

2.4 Classifier

We use a one-dimensional convolutional encoder followed by a single transformer encoder layer (Table 2). The strain enters as a length-16384 vector after whitening; three multi-scale convolutional stages (kernel sizes 7, 31, 63, fused 1×1) reduce the sequence length by a factor of 64, after which a single transformer encoder layer with $d_{\text{model}} = 32$, two heads, and feedforward dimension 64 models temporal correlations. The output is global-mean-pooled and projected through a two-layer MLP head to two class logits. Total trainable parameters: 1.73×10^5.

The classifier is trained with cross-entropy loss, AdamW ($\eta = 3 \times 10^{-4}$, weight decay 10^{-4}), batch size 32, and a cosine-annealed learning-rate schedule. For each $(\lambda_g, \rho_{\text{opt}})$ cell we generate a balanced set of 800 examples (400 per class), split 85%/15% into training and validation partitions (680 training, 120 validation), and train a separate classifier from scratch for 12 epochs. Reported metrics are always evaluated on the held-out validation partition of the same cell.

2.5 Evaluation

For both classifiers we report the area under the receiver–operator characteristic curve (ROC AUC) on the held-out validation partition. Bootstrap 90% confidence intervals on AUC are obtained from 1000 resamples with replacement of the validation set.

3 Implementation validation

We verify the dispersion implementation by computing the normalised match $\mathcal{M}(\tilde{h}_{\text{GR}}, \tilde{h}_{\text{MG}})$, maximised over time and phase, on H1 strain near GW150914 for a GW150914-like template ($m_1 = 36\,M_\odot, m_2 = 29\,M_\odot, D_L = 410\,\text{Mpc}$) and a logarithmic scan in λ_g from 10^{13} km to 10^{18} km. The mismatch $1 - \mathcal{M}$ crosses the detectability threshold $1/(2\rho_{\text{opt}}^2)$ at $\lambda_g \approx 10^{16}$ km for an event with $\rho_{\text{opt}} = 35.1$ (Fig. 1). This is consistent with the LIGO–Virgo–KAGRA O3 constraint $\lambda_g > 1.4 \times 10^{16}$ km [3], given that the published bound is obtained from an effective network SNR roughly twice this value through catalogue stacking.

4 Results

4.1 Sensitivity on observed strain

Figure 2 shows the validation AUC of both discriminators as a function of λ_g at the two target SNRs. The open squares are the signal-only control: with no strain noise added, the matched-filter statistic s_{MF} separates the GR and massive-graviton classes perfectly (AUC = 1.000) at every scanned λ_g, because the sign of the log-likelihood ratio is then noise-free. This confirms that the discriminator and the injection pipeline are free of implementation errors.

Once the same injections are embedded in observed H1 strain, the picture changes completely. The oracle matched filter gives AUC = 0.46–0.49 and the convolutional–transformer classifier AUC = 0.45–0.48, the two differing by at most ≈ 0.02 at any cell. Every bootstrap 90% confidence interval includes the chance value 0.5, so neither method achieves a statistically significant single-event detection at any $(\lambda_g, \rho_{\text{opt}})$ on the grid, and the two methods are indistinguishable from each other. Increasing the injected SNR from 20 to 50, or decreasing λ_g (which deepens the dispersion signature), does not lift either method off the chance line.

4.2 ROC at the most discriminative cell

Figure 3 compares the full ROC curves at the most favourable cell on the grid—the smallest $\lambda_g = 10^{14}$ km at the higher SNR $\rho_{\text{opt}} = 50$, where the signal-only template mismatch is largest.

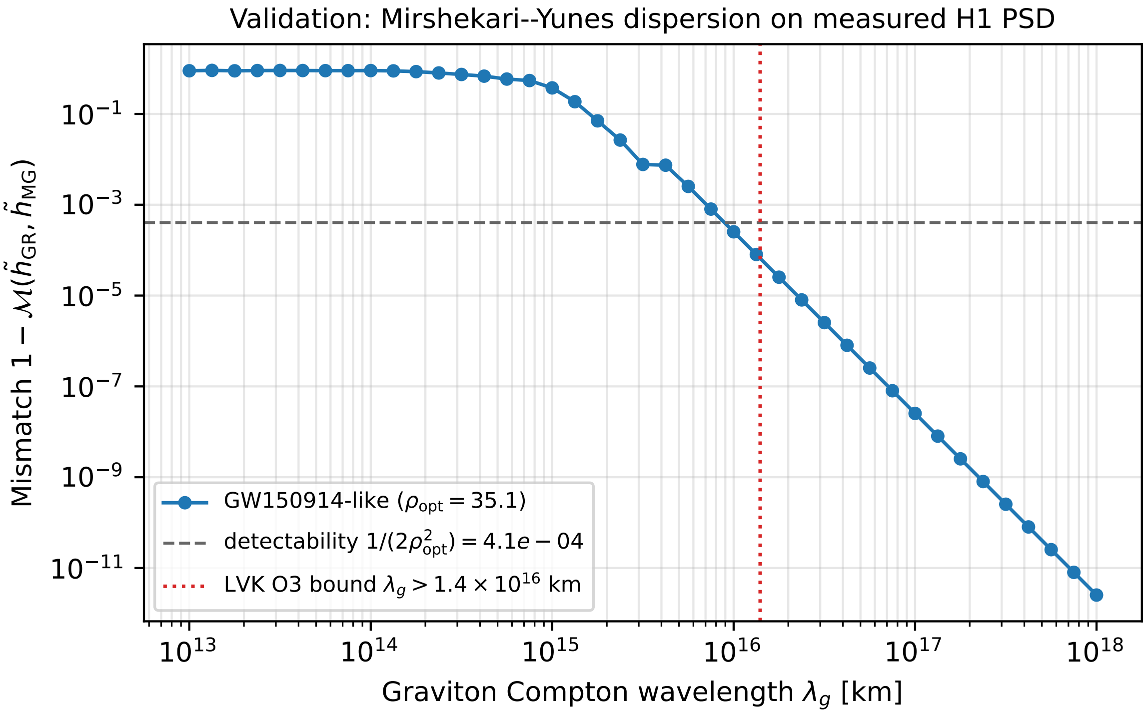

Figure 1: Mismatch $1 - \mathcal{M}(\tilde{h}_{\rm GR}, \tilde{h}_{\rm MG})$ as a function of λ_g for a GW150914-like template at $D_L = 410\,$Mpc, evaluated on the same 1024-second off-source H1 PSD used for the injection study (Sec. 2.2). Dashed line: per-event detectability threshold $1/(2\rho^2_{\rm opt})$ for the optimal SNR $\rho_{\rm opt} = 35.1$ of this template. Dotted line: the LIGO–Virgo–KAGRA O3 90% upper limit $\lambda_g > 1.4 \times 10^{16}\,$km [3].

Even here both curves track the chance diagonal, and their bootstrap 90% confidence bands overlap each other and enclose the diagonal along its entire length. The matched filter and the learned classifier therefore deliver the same (null) single-event performance at the point where any dispersion signature should be easiest to detect.

4.3 Positive control: the classifier learns when the noise is removed

The null result of Sec. 4.1 could in principle reflect a network too small, or a training set too sparse, to learn the dispersion signature at all. Figure 4 rules this out. When the identical injections are presented without strain noise (*signal only*), the same compact architecture, trained with the same optimiser and schedule, separates the GR and massive-graviton classes almost perfectly: at $N = 400$ examples per class the validation AUC climbs from chance to $\gtrsim 0.94$ within a few epochs [Fig. 4(a)], and the best validation AUC approaches unity at every training-set size tested [Fig. 4(b)]. On H1 strain, by contrast, the AUC of the same network never departs from chance, at any epoch or any N. The network therefore has ample capacity to detect the inter-class difference; what defeats it on strain is the detector noise, not a lack of model capacity or training data. Increasing N from 100 to 400 per class does not lift the strain AUC, showing that single-event sensitivity has already saturated with respect to training-set size in this regime—the classifier analogue of the noise-free matched-filter control of Fig. 2.

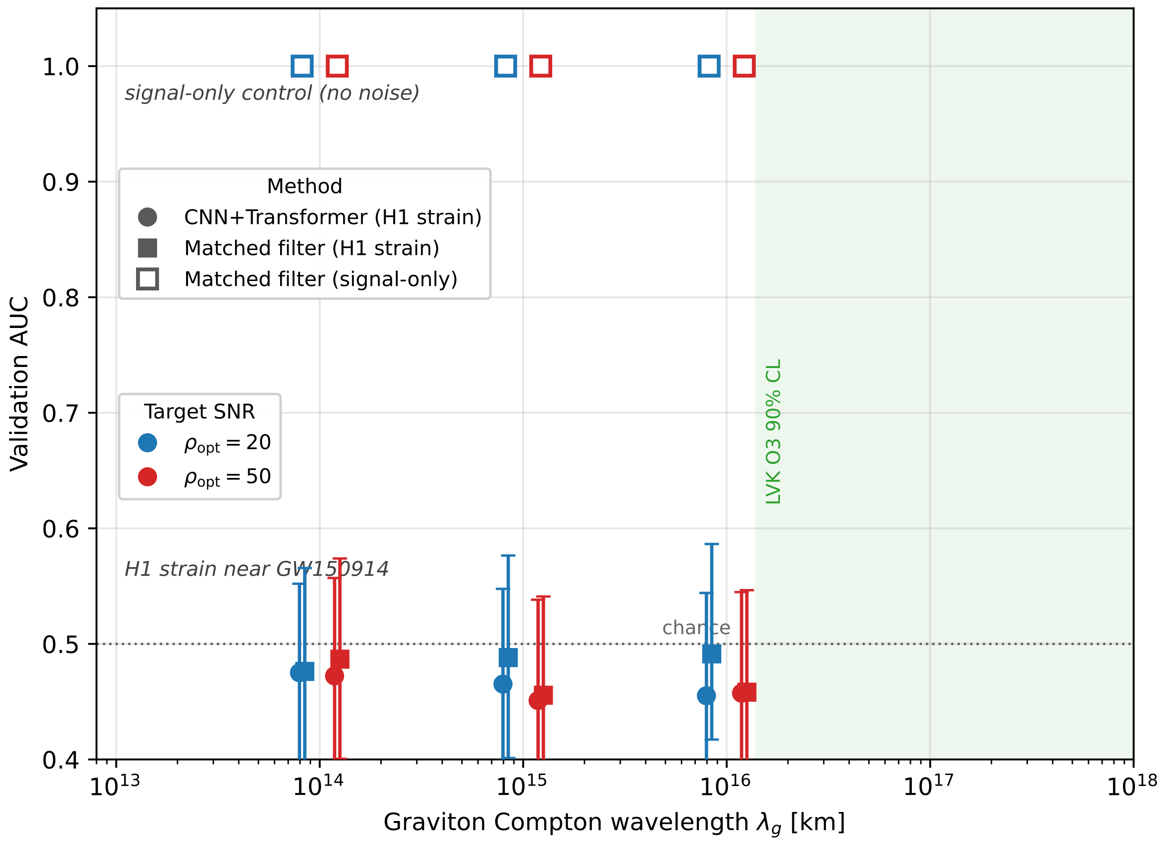

Figure 2: Single-event AUC for the oracle matched filter (squares) and the convolutional–transformer classifier (circles) as a function of λ_g at two target optimal SNRs. Open squares: matched-filter signal-only control (no strain noise added), pinned at unity because the sign of the log-likelihood ratio is noise-free. Filled symbols: H1 strain near GW150914. Error bars: bootstrap 90% confidence intervals on AUC (1000 resamples). Vertical band: LIGO–Virgo–KAGRA O3 90% upper limit on λ_g [3].

5 Discussion

The per-event AUC of the oracle matched filter on H1 strain hovers near 0.46–0.49 across the entire scanned $(\lambda_g, \rho_{\rm opt})$ grid. We have ruled out the possibility that this reflects an implementation defect: on signal-only data the same code returns AUC compatible with unity at all $\lambda_g \leq 10^{16}\,$km, and the per-event detectability threshold computed from Eq. (3) reproduces the catalogue-stacked O3 bound to within the factor expected from network SNR scaling (Sec. 3). What dominates the per-event discriminator on strain is the variance contributed by non-Gaussian features in the data — blip glitches [11, 25], harmonic lines, and short-duration transients — whose energy partially projects onto both the GR and the dispersed templates, producing correlated fluctuations in $\rho^2_{\rm GR}$ and $\rho^2_{\rm MG}$ that do not cancel in $s_{\rm MF}$ because the two templates have slightly different time–frequency support.

Figure 5 makes this scale separation explicit. Panel (a) shows the noise-free whitened GR and massive-graviton templates for a GW150914-like source at $\rho_{\rm opt} = 20$, together with their difference $h_{\rm MG} - h_{\rm GR}$ on the same vertical scale. Even at the deepest dispersion on our grid ($\lambda_g = 10^{14}\,$km), the entire inter-class signal that any single-event discriminator must detect peaks at only ≈ 1.7 noise standard deviations. Panel (b) shows a 4-second segment of off-source H1 strain with the identical injection added, whitened on the same scale: the data are pervasively non-Gaussian, the loudest localised transient reaching $\approx 8\sigma$ on a baseline already several times the inter-class difference. It is this excess, and not any deficiency of the discriminator, that drives the single-event AUC to chance.

The convolutional–transformer classifier learns no representation that appreciably reduces this glitch-induced variance on the per-event task, at the training-set scales and SNRs studied here. This is consistent with the LIGO–Virgo–KAGRA practice of obtaining graviton-mass constraints by stacking $\mathcal{O}(20)$ events; the present findings would not predict a useful per-event constraint from any single-event learned discriminator at the SNRs of the GWTC-4.0 catalogue.

Limitations. (i) The noise reservoir is restricted to a single $\sim$ 17-minute segment of O1 H1 strain near GW150914. Other observing runs and detectors carry different glitch populations; the sensitivity numbers do not generalise across runs without further analysis. (ii) The matched filter is taken in oracle form (templates at the true source parameters). A practical bank-template search [19] or stochastic-sampling parameter estimation [26] would exhibit modestly worse sensitivity than the oracle, so the classifier-versus-MF comparison reported here is conservative against the classifier. (iii) We use leading-order Mirshekari–Yunes dispersion only; subleading effects on merger and ringdown morphology [27] are not modelled. (iv) The training set size ($N = 800$ examples per cell) is small by GW deep-learning standards [14, 13]; the training-set-size study of Fig. 4(b) shows the single-event sensitivity already saturated with respect to N in this regime, because the per-event variance is set by the strain rather than by the training-set size. We nonetheless cannot exclude modest gains from substantially larger datasets or a wider sampling of the noise. (v) The CNN+Transformer architecture is one of many possible choices; specialised models trained on glitch-rich data (*e.g.*, transient-aware or attention-based architectures [28]) may give different results, and post-hoc attribution methods [29] could localise which time–frequency features a successful discriminator exploits. We do not claim our architecture is optimal.

6 Conclusion

We set out to test, on a *single* gravitational-wave event, the general-relativistic prediction that the graviton is massless, and to ask whether a learned discriminator can detect a massive-graviton dispersion signature more sensitively than the optimal matched filter. Using balanced injections of general-relativistic and Mirshekari–Yunes–dispersed inspirals—source parameters drawn from the GWTC-4.0 catalogue, dispersion phase implemented to leading order in $1/\lambda_g^2$—into H1 strain near GW150914, we evaluated a compact convolutional–transformer classifier and the oracle matched filter on identical examples. The two methods are statistically indistinguishable at every $(\lambda_g, \rho_{\text{opt}})$ cell scanned, each saturating in the range AUC = 0.45–0.49 with confidence intervals that include chance. Two controls confirm this is a property of the data rather than an implementation defect: the pipeline reproduces the published O3 graviton-mass bound to within the factor expected from network-SNR scaling [3], and on noise-free signals the same matched filter separates the two waveform families perfectly at every scanned λ_g.

The single-event sensitivity is therefore set by non-Gaussian detector noise—blip glitches and short-duration transients—and not by the choice of discriminator: at the signal-to-noise ratios typical of the current catalogue, a learned representation offers no shortcut around the noise floor that obliges LIGO–Virgo–KAGRA to stack $\mathcal{O}(20)$ events for a useful graviton-mass constraint [10]. This is a null methodological result, and it delimits where future gains must originate. A per-event advantage for deep learning, if one exists, would most plausibly emerge from representations trained explicitly to be robust against glitches, from models that learn to combine many events coherently rather than scoring each in isolation, or from the substantially higher single-event SNRs anticipated with detector upgrades and next-generation observatories. The framework introduced here—open injections, an oracle matched-filter baseline, and a like-for-like classifier comparison on identical data—provides a reusable benchmark against which any future claim of a learned single-event advantage can be tested.

Data and code availability

All code, the curated GWTC-4.0 event list, the result data and figures, and the bibliography are available at `https://github.com/ramc77/gw-graviton-ml-benchmark`. The GW strain data are public via the LIGO Open Science Center, `https://gwosc.org`.

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A Curated GWTC-4.0 event list

Table 1: Source-frame masses, luminosity distance, and network matched-filter SNR for the GWTC-4.0 BBH events used as the parameter pool for our injection campaign ($m_1, m_2 > 3\,M_\odot$; network SNR > 10). Top 12 by SNR shown; full list (56 events) in `gwtc4_bbh_events.csv` in the accompanying code repository. Values from the `gwosc.api` event JSON endpoint.

Event	$m_1\,(M_\odot)$	$m_2\,(M_\odot)$	D_L (Mpc)	ρ
GW230814_230901	33.7	28.2	290	43.0
GW231226_101520	40.2	35.1	1180	34.7
GW230627_015337	8.4	5.7	305	28.7
GW231028_153006	94.0	59.0	4200	22.4
GW231206_233901	37.6	28.5	1490	21.9
GW231123_135430	137.0	101.0	2200	21.8
GW230927_153832	21.7	16.6	1190	20.3
GW230914_111401	60.0	37.0	2700	17.2
GW230919_215712	27.3	21.4	1330	16.8
GW230628_231200	32.5	27.0	2260	16.4
GW231102_071736	61.0	42.0	3800	15.6
GW240104_164932	42.3	32.2	1910	14.8

B Classifier architecture

Table 2: Layers and parameter counts in the compact convolutional plus transformer classifier. Total trainable parameters: 1.73×10^5.

Layer	Output / parameters
Input strain	$(B, 1, 16384)$
Multi-scale conv stage 1 (kernels 7, 31, 63)	$(B, 12, 4096)$
Multi-scale conv stage 2	$(B, 24, 1024)$
Multi-scale conv stage 3	$(B, 48, 256)$
1×1 projection to $d_{\text{model}} = 32$	$(B, 32, 256)$
Sinusoidal positional encoding	—
Transformer encoder layer (2 heads, FF = 64)	$(B, 256, 32)$
Global mean pool	$(B, 32)$
MLP head ($32 \to 64 \to 2$)	$(B, 2)$

C Mirshekari–Yunes phase derivation, units convention

The dispersion relation for a massive graviton of rest mass m_g gives a group velocity $v_g/c = \sqrt{1-(mc^2/E)^2}$ with $E = hf$. Expanding to leading order in $(c/\lambda_g f)^2$ gives a time delay across luminosity distance D_L of $\Delta t(f) = D_L c/(2\lambda_g^2 f^2)$, and the resulting phase shift in the matched-filter convention of Eq. (3) is [2]

$$\delta\Psi_{\text{MG}}(f) = -\frac{\pi^2\, c\, D_L}{\lambda_g^2 (1+z)\mathcal{M}_c\, f}, \tag{5}$$

with D_L in meters, λ_g in meters, $\mathcal{M}_c = G M_c/c^3$ in seconds, f in hertz, and $c = 2.998 \times 10^8$ m/s. The factor of $\mathcal{M}_c^{-1}$ comes from absorbing the dispersion into the post-Newtonian expansion at order $u^{-3} = (\pi \mathcal{M}_c f)^{-1}$ and is what gives the formula the correct order of magnitude (micro-radian per cycle scales rather than micro-microradian); omitting it yields a phase modification four orders of magnitude too small to reproduce the published O3 bound.

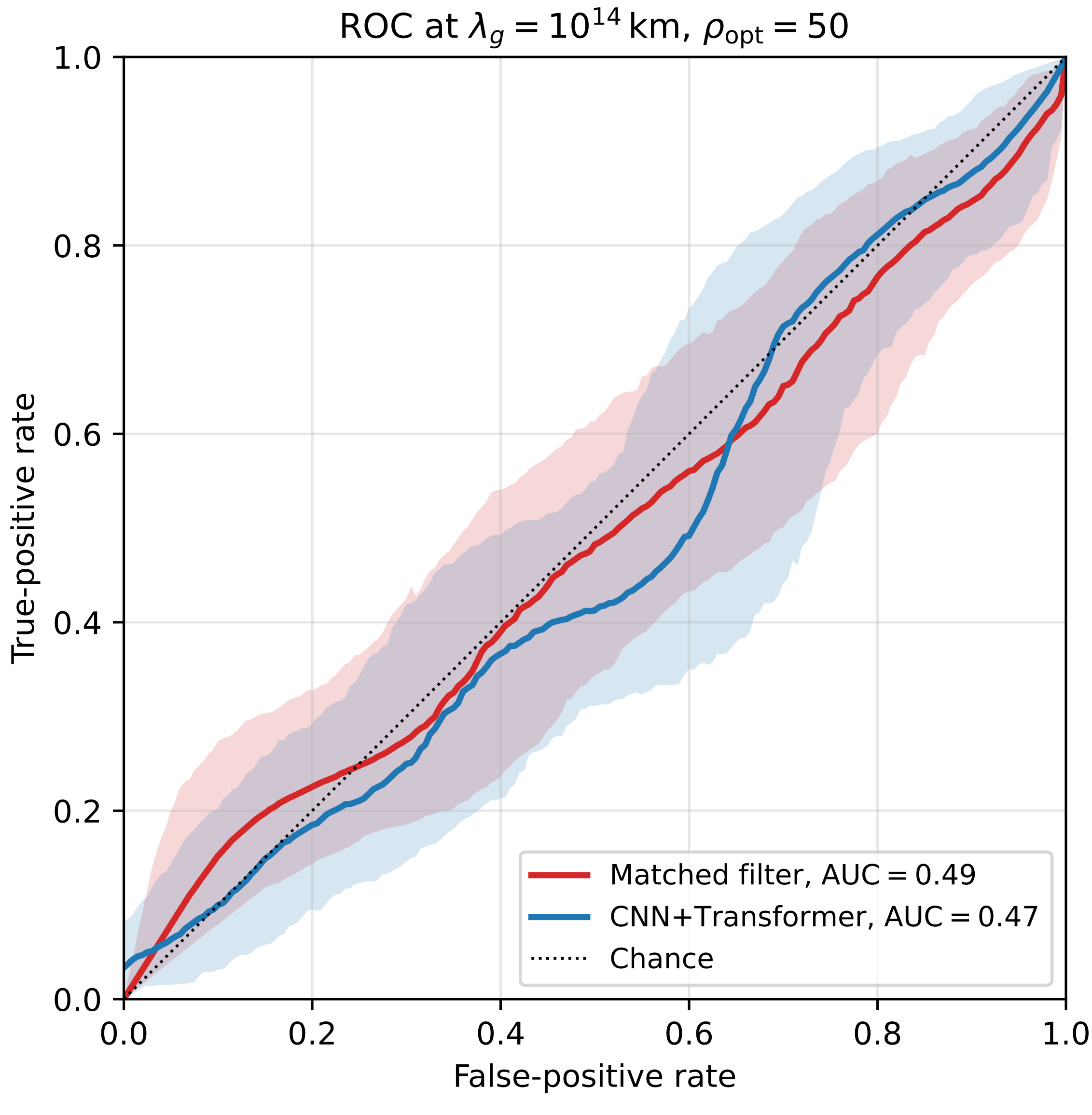

Figure 3: ROC at the cell with the largest signal-only mismatch in the scanned grid. Shaded bands: bootstrap 90% confidence regions.

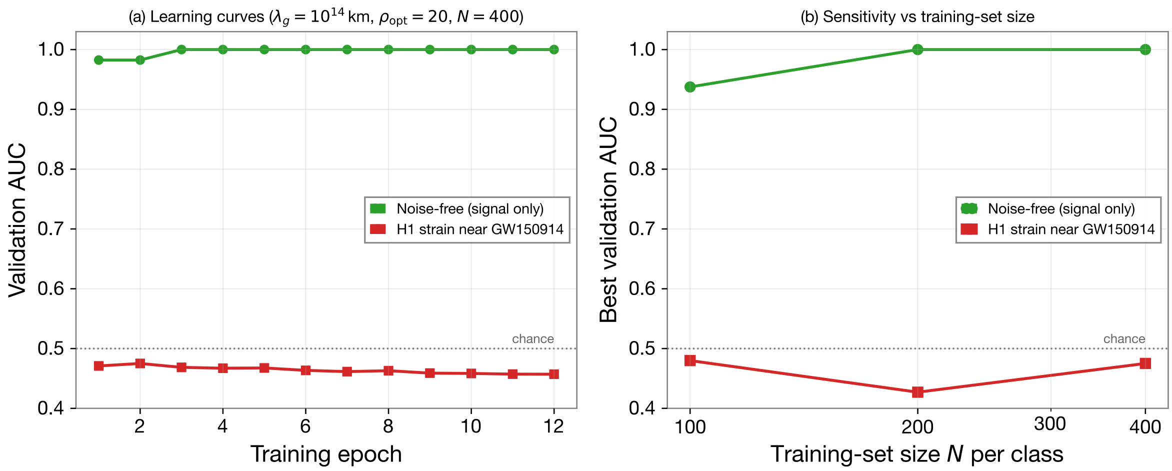

Figure 4: Classifier positive control at the most discriminative cell ($\lambda_g = 10^{14}$ km, $\rho_{\rm opt} = 20$). (a) Validation AUC versus training epoch at $N = 400$ examples per class: noise-free (signal-only) injections (circles) versus H1 strain near GW150914 (squares). (b) Best validation AUC versus training-set size N per class. Noise-free, the network separates the classes at every N; on strain it remains at chance (dotted line). The contrast establishes that the null result is set by the detector noise, not by network capacity or training-set size.

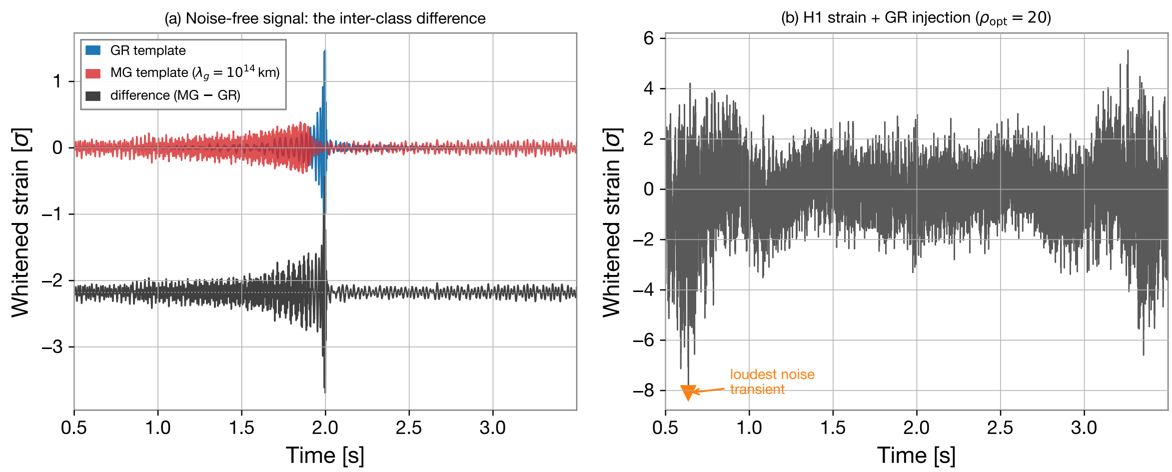

Figure 5: Why single-event discrimination is noise-limited. (a) Noise-free whitened GR template, massive-graviton template ($\lambda_g = 10^{14}$ km), and their difference (offset below, same vertical scale) for a GW150914-like source at $\rho_{\rm opt} = 20$; the inter-class difference peaks at $\approx 1.7\sigma$. (b) A 4-s off-source H1 strain segment near GW150914 with the identical GR injection added, whitened on the same noise scale; the loudest localised noise transient (marked) and the pervasively non-Gaussian baseline dwarf the inter-class difference of panel (a). Both panels are in common units of detector-noise standard deviations.